Interop QOS

2026-03-23-12-11-52

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions. By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels, ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

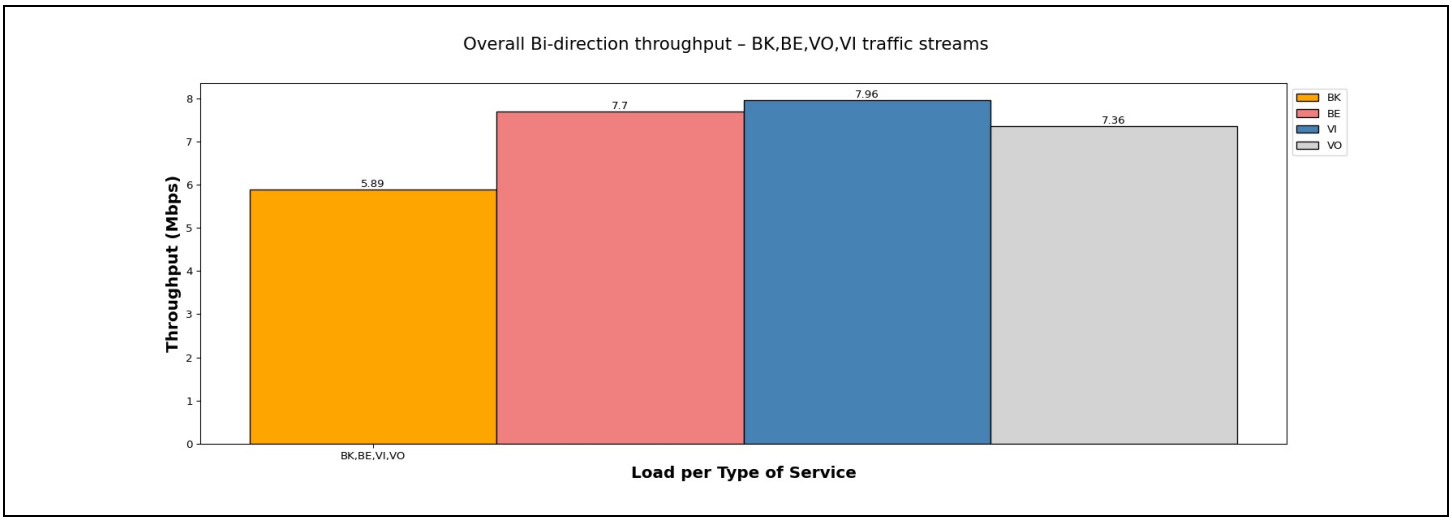
Test Configuration	Device List	vivonew(Android), sungsung(Android), ubuntu24(Linux), macoss-MacBook-Air.local(Mac)
	Number of Stations	Total(4) Android(2) Linux(1) Mac(1)
	AP Model	Cisco
	SSID	NETGEAR_5G_wpa2
	Traffic Duration in hours	0.02
	Security	wpa2
	Protocol	UDP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:1.0 Mbps,Download:1.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-upload	Throughput for Load 1.0 Mbps-download
4	BK : 2.94, BE : 3.85, VI: 3.98, VO: 3.67	BK : 2.95, BE : 3.85, VI: 3.98, VO: 3.69

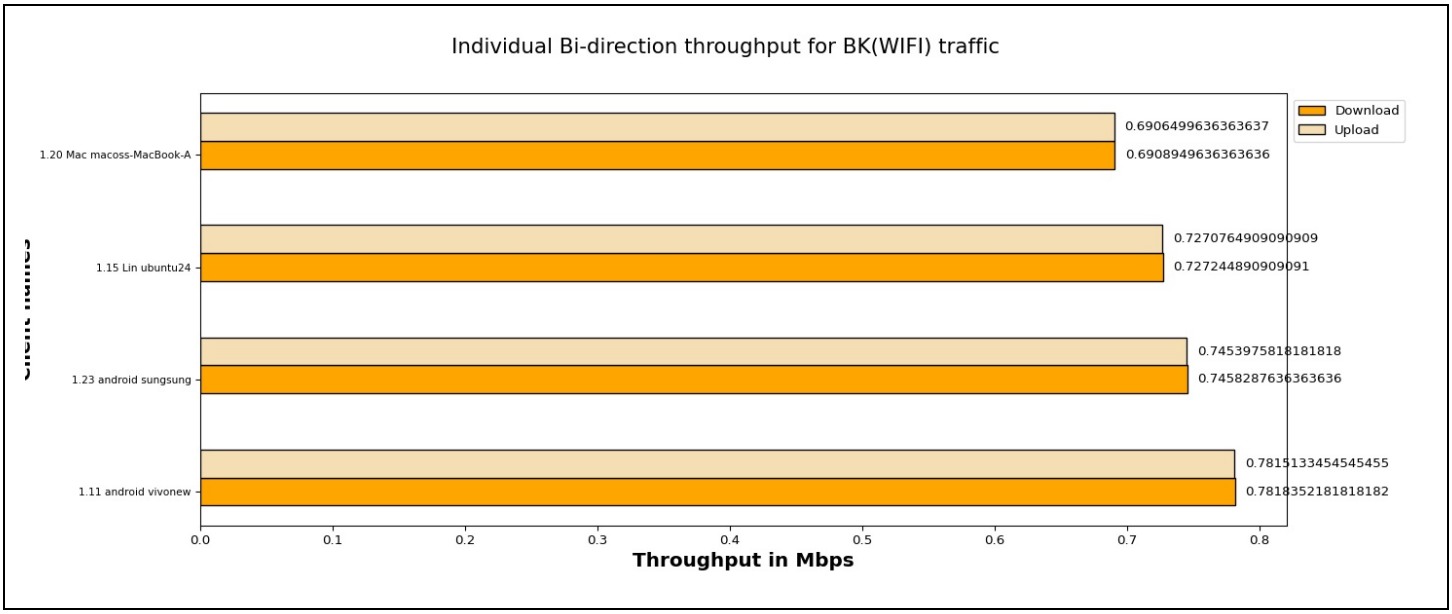
Overall Bi-direction throughput for 4 clients with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loadsUpload:1.0 Mbps,Download:1.0 Mbps per tos



Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BK(WiFi).

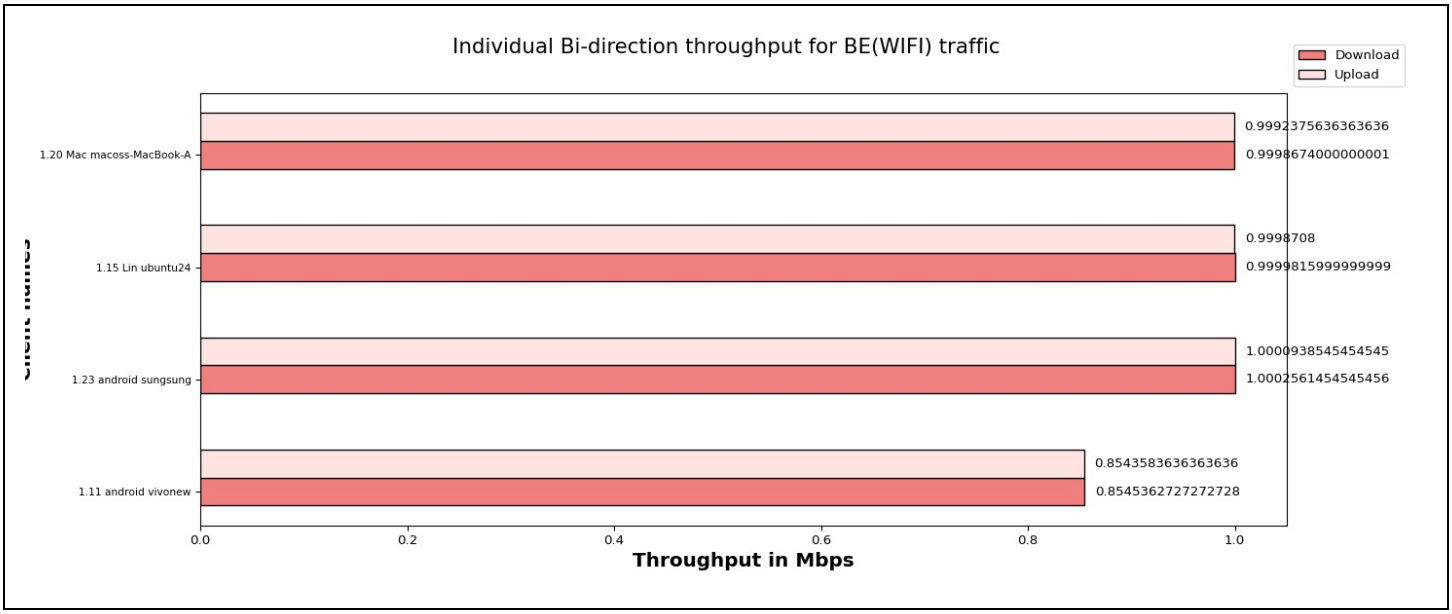
The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7815133454545455 Mbps	0.7818352181818182 Mbps	0.000497	0.011014
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7453975818181818 Mbps	0.7458287636363636 Mbps	0.000000	0.002198
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7270764909090909 Mbps	0.727244890909091 Mbps	0.003888	0.007883
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.6906499636363637 Mbps	0.6908949636363636 Mbps	0.008233	0.003604

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".

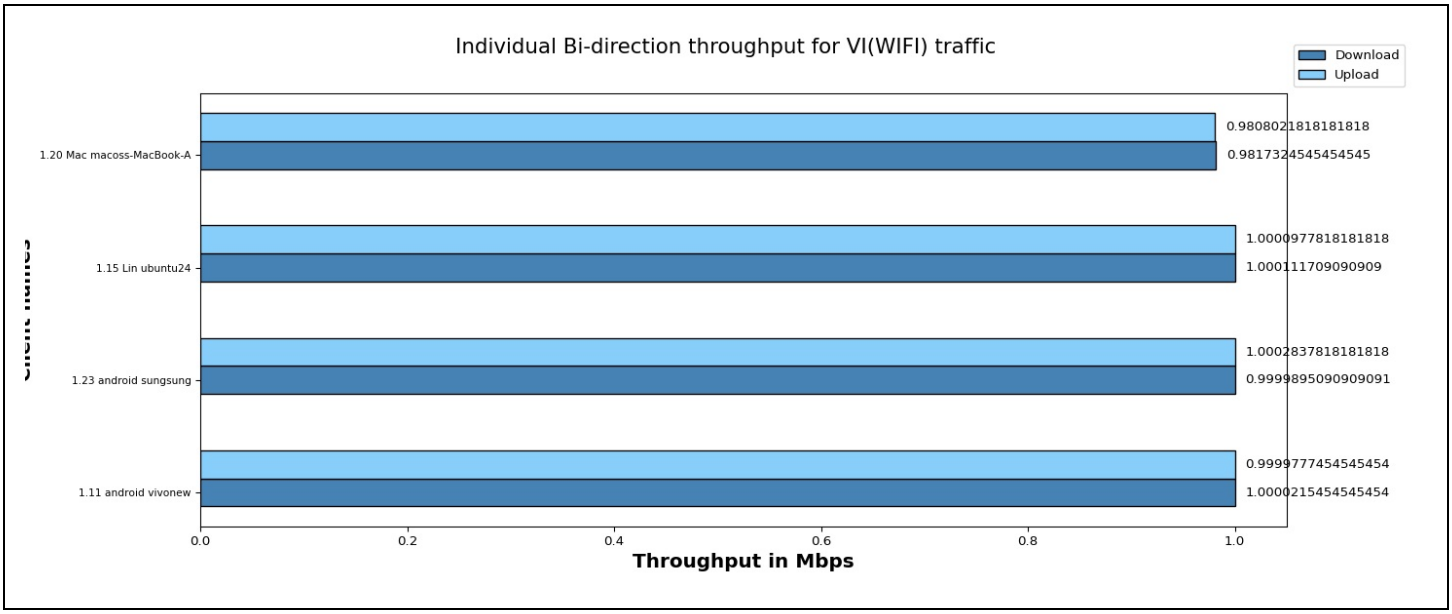


Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.8543583636363636 Mbps	0.8545362727272728 Mbps	0.001581	0.003836

1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	1.0000938545454545 Mbps	1.0002561454545456 Mbps	0.002973	0.000917
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.9998708 Mbps	0.9999815999999999 Mbps	0.009013	0.007223
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.9992375636363636 Mbps	0.9998674000000001 Mbps	0.000000	0.002122

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

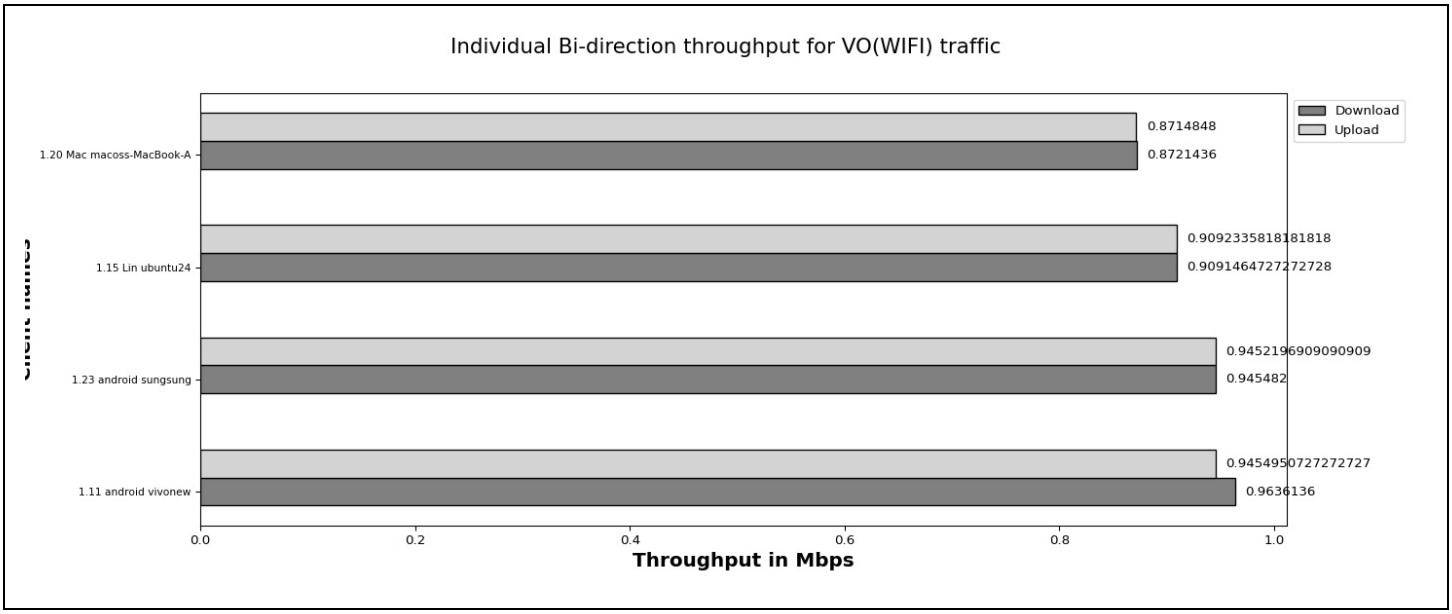
The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.9999777454545454 Mbps	1.0000215454545454 Mbps	0.000637	0.000674
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	1.0002837818181818 Mbps	0.9999895090909091 Mbps	0.004085	0.000000
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	1.0000977818181818 Mbps	1.000111709090909 Mbps	0.000000	0.008390
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.9808021818181818 Mbps	0.9817324545454545 Mbps	0.015762	0.004626

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9454950727272727 Mbps	0.9636136 Mbps	0.001374	0.000000
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9452196909090909 Mbps	0.945482 Mbps	0.004884	0.001168
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9092335818181818 Mbps	0.9091464727272728 Mbps	0.003279	1.958302
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.8714848 Mbps	0.8721436 Mbps	0.000000	0.001912

Information	contact	support@candelatech.com